Annu. Rev. Earth Planet. Sci. 1990. 18:257–86
Copyright © 1990 by Annual Reviews Inc. All rights reserved

ANNUAL REVIEWS Further
Quick links to online content

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

SEISMIC DISCRIMINATION OF NUCLEAR EXPLOSIONS

Paul G. Richards

Lamont-Doherty Geological Observatory and Department of Geological Sciences, Columbia University, Palisades, New York 10964

John Zavales

School of International and Public Affairs, Columbia University, New York, New York 10027[1]

INTRODUCTION

The monitoring of underground nuclear explosions has been discussed for over 30 years in connection with verification of test ban treaties, and monitoring is also one of the basic ways to learn about nuclear weapons development in different countries. Explosion monitoring is thus an important technical exercise, occasionally attracting attention from high-level policymakers and generating front-page news. We can expect this level of interest to continue as long as there is debate on the merits of modernizing nuclear weapons.

The underlying issues here are related to the uses of nuclear weapons, their cost, and the strategic capabilities and vulnerabilities of the superpowers and other countries. Because these are some of the largest and most contentious questions of our time, arguments sometimes spill over from the policy arena into the technical subject of monitoring capability itself.

As a side effect of the debate, geophysics (especially seismology) continues to be shaped by significant funding for research and development in verification/monitoring technologies. New funds at the level of about

[1] Now at US Department of the Army, Security Affairs Command, Alexandria, Virginia 22332.

0084–6597/90/0515–0257$02.00

$30 million came into seismic monitoring programs in the United States in 1989.

In the sections that follow we review the history of nuclear testing and test ban negotiations, introduce some basic terminology, give examples of detection capability, describe characteristics of different seismic sources and methods of explosion identification, and briefly review different evasion scenarios. We conclude with a summary statement of identification capability and its implication for future possibilities in verifying new nuclear testing limitations.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

BRIEF HISTORY OF NUCLEAR TESTING AND NEGOTIATIONS

The rate at which nuclear explosions have been conducted is characterized in Figure 1 by a pair of histograms giving the numbers carried out each year by the superpowers, beginning in 1945 with TRINITY. The World War II explosions in Japan at Hiroshima and Nagasaki are included. Atmospheric explosions are shown plotted above the zero line, and underground explosions are plotted below.

Note the extensive series of atmospheric tests in the 1950s and again in the early 1960s. The first underground nuclear explosion, code-named RAINIER, was carried out in 1957 in the United States at the Nevada Test Site. The gap for the years 1959 and 1960 represents the moratorium on testing that began in 1958 and ended in 1961, during a time of severe antagonism between the US and the USSR over nuclear testing issues, when the Soviets quite suddenly began a major atmospheric test program. In 1963 this type of testing was banned by the Limited Test Ban Treaty (LTBT), which prohibited nuclear explosions in the atmosphere, oceans, and space, but permitted them underground.

Since 1963, the US and the USSR have each carried out about 10 to 30 underground nuclear explosions per year. More recently, a unilateral moratorium was announced by the Soviet Union, which began in 1985 and ended in 1987 (Figure 1).

The main Soviet sites for nuclear weapons testing are at Novaya Zemlya (an island north of the Eurasian land mass) and in eastern Kazakhstan, near the city of Semipalatinsk. The eastern Kazakhstan site includes both Degelen and Shagan River as separate test sites. All explosions of concern with respect to the 150-kt Threshold Test Ban Treaty (TTBT) have occurred since 1976 at Shagan River. [A kiloton (kt) is a unit of energy used to quantify the yield of a nuclear explosion. Originally defined as the energy equivalent to exploding a thousand tons of TNT, it is now defined as 10^{12} cal.] The USSR has also set off about 100 peaceful nuclear

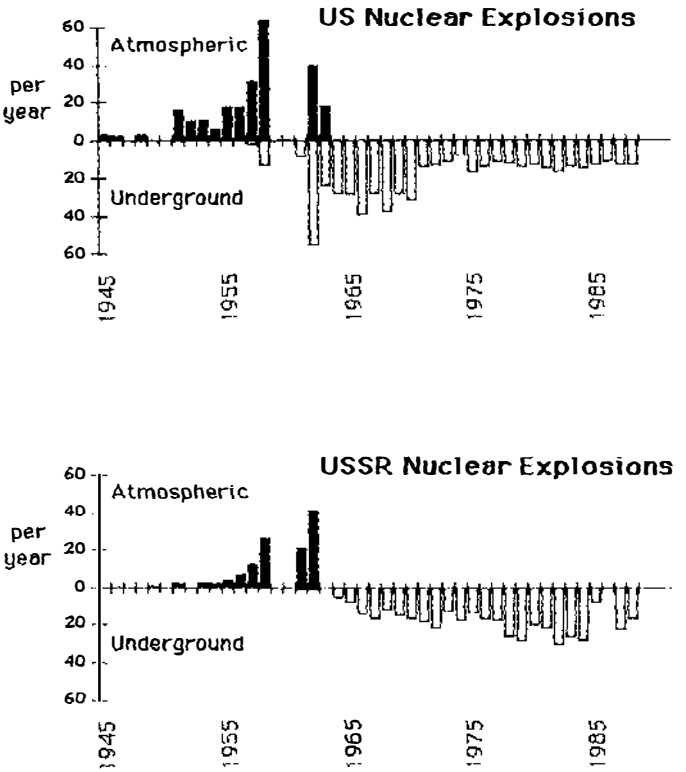

Figure 1 Numbers of nuclear explosions conducted each year by the US and USSR to 1988. These numbers include explosions announced as nuclear by the nation conducting the test, plus explosions routinely identified as "presumed nuclear" by organizations such as the US Geological Survey and the Swedish Ministry of Defence.

explosions, away from weapons test sites, for a variety of purposes that include creation of underground cavities for storage of chemicals, putting out oil-well fires, damming a river, and providing seismic signals for geophysical surveys of the structure of the continent (Scheimer & Borg 1984). The US now restricts nuclear testing to the Nevada Test Site, though in the past a few underground explosions were carried out elsewhere in the western US, in Mississippi, and in the Aleutian Islands of Alaska.

Underground nuclear explosions were conducted in the Sahara, in what is now part of Algeria, by France in the 1960s. In recent years the French

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

have continued testing underground at sites in the South Pacific, notably on the Mururoa Atoll in the Tuamoto Archipelago. The United Kingdom has a test program, now conducted in the US at the Nevada Test Site. The Chinese have also tested nuclear weapons underground, with explosions at Lop Nor in the Xinjiang region of northwest China, and India carried out one underground nuclear explosion in 1974.

Nuclear testing was one of the leading international political issues of the 1950s and 1960s. During the presidencies of Dwight Eisenhower and John Kennedy, negotiations were conducted with the Soviet Union for a ban on underground tests above seismic magnitude 4.75 (associated at the time with tests above about 20 kt TNT equivalent), together with a moratorium on smaller underground tests. A joint seismic research program with the Soviets was also proposed in order to improve the monitoring of these smaller tests so that they too could eventually be prohibited in a Comprehensive Test Ban Treaty (Seaborg 1981). But, ostensibly because of disagreements regarding on-site inspections, how the operation of seismic monitoring would be conducted, and who would control the organization that carried out the monitoring, the treaty that eventually emerged in 1963 placed no limits whatsoever on testing underground, other than a ban on nuclear explosions that cause radioactive debris to cross national borders. As a part of the political process of securing the advice and consent of the US Senate, President Kennedy made a commitment, under the so-called safeguards associated with the LTBT, to promote a vigorous program of underground nuclear testing.

Widespread interest in nuclear testing resumed during the presidency of Richard Nixon, when quite suddenly in 1974 the Threshold Test Ban Treaty (TTBT) was negotiated and signed at a summit meeting in Moscow. This treaty introduced a new technical issue of importance in treaty monitoring, namely the question of yield estimation, because it banned underground nuclear weapons tests greater than 150 kt. In addition to the requirements of detecting these explosions and of identifying them as nuclear, it now became important to estimate their size. Some would say the TTBT is a fairly minor arms control treaty, but insofar as it was intended to terminate the practice of high-yield testing, it can be called successful. Both superpowers had been carrying out weapons tests prior to 1974 at yield levels in excess of 1 Mt. Though still not ratified as of 1989, this treaty is in effect and has been the focus of political debate on nuclear testing in both the Reagan and Bush administrations. The companion to the TTBT is the Peaceful Nuclear Explosions Treaty (PNET) of 1976, which was concluded after 18 months of intensive discussion and which has a very extensive protocol for verification. This treaty bans nonmilitary underground nuclear explosions that individually

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

exceed 150 kt, though in certain circumstances, and subject to special in-country monitoring, it permits a salvo of nuclear explosions with total yield exceeding the TTBT limit.

The Carter administration made a commitment to achieve a comprehensive test ban. In trilateral negotiations (1977–80) that included the United Kingdom, agreement was eventually reached on the monitoring network of seismometers to be deployed in the US and USSR, on allowing an unlimited number of "challenge-type" on-site inspections, and on a moratorium for peaceful nuclear explosions. Acceptance of a request for on-site inspection was not to be mandatory, but it was agreed that the rejection of a request would be a serious matter (see Heckrotte 1988). The monitoring of those treaties now in effect (LTBT, TTBT, PNET) does not require access to "in-country" seismic data because seismic signals for explosions approaching the 150-kt threshold are large enough to be picked up by seismometers all over the globe. However, for a Comprehensive Test Ban Treaty (CTBT) or a Low Yield Threshold Test Ban Treaty (LYTTBT), in-country data would be essential. The Carter administration's test ban negotiations were eventually held up by debate over larger political issues (such as SALT II), became mired in detail over the number of seismic stations required on United Kingdom territory, and never reached their intended goal (York 1987, Chap. 14).

On July 20, 1982, early in the Reagan administration, a press briefing was given at the White House announcing that trilateral test ban negotiations would not resume, and that there were problems with verification of the TTBT and PNET. Specifically, it was stated that "on several occasions, seismic signals from the Soviet Union have been of sufficient magnitude to call into question Soviet compliance with the threshold of 150 kt, and it's because of the uncertainties of our yield estimation process that the United States cannot prove beyond any reasonable doubt that the Soviets have violated the Threshold Test Ban Treaty. . . . The Soviets have always asserted, when challenged on this point, that they have not violated the agreement." Subsequently, the Reagan administration stated that a comprehensive treaty remained a long-term goal of the United States, but provided a series of restrictions governing the circumstances under which a CTBT would be favored. The main issue concerning nuclear testing in the Reagan years became the assessment of Soviet compliance with the 150-kt yield threshold, with the Administration reporting several times to the US Congress that "while, in view of ambiguities in the pattern of Soviet testing and in view of verification uncertainties, the available evidence is ambiguous and we have been unable to reach a definitive conclusion, this evidence indicates that Soviet nuclear-testing activities for a number of tests constitute a likely violation of legal obligations under the TTBT."

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

Such claims became contentious because no technical evidence was publicly offered in support, and those scientists who had presumably participated in governmental review of the evidence typically reached a different conclusion. For example, the Director of the Lawrence Livermore National Laboratory, with responsibilities for designing and certifying US nuclear warheads, stated in 1987 (Batzel 1987) that "the Soviets appear to be observing some yield limit. Livermore's best estimate of this yield limit . . . is that it is consistent with TTBT compliance. However . . . , we cannot rule out the possibility that a few Soviet tests may have exceeded the limit."

Between 1986 and 1988, the US Congress threatened several times to cut off funds (amounting to about $800 million per year) for the US nuclear test program, and in September 1987 a US-Soviet joint statement announced the resumption of negotiations on limitations on nuclear testing with the following words: "In these negotiations the sides as the first step will agree upon effective verification measures which will make it possible to ratify the US-USSR Threshold Test Ban Treaty of 1974 . . . and [will] proceed to negotiating further intermediate limitations on nuclear testing."

OVERVIEW OF TECHNICAL ISSUES

There are three main steps in seismic monitoring of underground nuclear explosions. The first is that of *detection*, usually taken to mean detection of a signal by enough stations to enable location of the seismic source. The second step is that of *identification*, the main technical subject under discussion in this review paper. And the third step is that of *yield estimation* (Richards 1988, Sykes & Ruggi 1989). Because of the technical problems introduced by the 1974 TTBT, most research in explosion monitoring in the years 1975–85 was concerned with achieving accurate and credible yield estimates for large explosions. (It is remarkable that the US and the USSR each sent substantial teams for the first time in 1988 to the other side's main weapons test site to improve techniques of yield estimation. Negotiations on improving verification of the TTBT and PNET continue under the Bush administration.) However, it appears that yield estimation for large explosions is no longer the most challenging frontier in monitoring research. Promoted in part by the superpowers' joint statement of 1987, the research focus has shifted back to the question of what identification capability for small explosions can be achieved by novel seismic instrumentation and data analysis.

Associated with each of the three main technical steps in explosion monitoring (detection, identification, yield estimation) is a set of questions about what can be done to evade, or "spoof," the side doing the monitor-

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

ing. Thus, one asks how well a nuclear explosion could be hidden in various environments. This debate has to be developed in terms of thresholds: What size nuclear explosion might go undetected; what size unidentified; and at what size can yield estimation be spoofed? Nuclear explosions can be conducted even down to such small sizes as pounds or even a few grams of TNT equivalent (Thorn & Westervelt 1987). It is becoming more widely recognized that there is an interaction between policy options and seismic monitoring capability that is driven by the following two technical points: (*a*) It is not possible to monitor a truly comprehensive test ban; and (*b*) a low-yield threshold test ban would require of monitoring programs the capability to estimate yield, which is increasingly difficult to do as yield level is reduced.

There is therefore a fundamental trade-off between the stringency of desired limitations on testing and the stringency of achievable verification standards. The monitoring objective in general is to drive downward the yield level at which even well-organized and well-funded attempts at clandestine testing might be successful. Looking at this another way, monitoring might be associated with some type of requirement to demonstrate that clandestine testing would be detected and identified if it occurs above a yield level that is significant from the perspective of weapons design.

Somewhat different conclusions can be reached on the three main steps in seismic monitoring regarding the quality of current capabilities. For example, it appears that detection capability in general is very good. Some examples to illustrate this capability are given in the next section. Current identification capability is also very good for events above a certain size, including those that raise questions of compliance with the 150-kt limit of the TTBT; is good even for events 10 times smaller; and is adequate in most but not all situations down to around 1 kt. No agreement exists, however, on how well nuclear explosions can now be identified at very low magnitude, and on whether the US could, with access to seismic data from the Soviet Union, identify all underground nuclear explosions down to 1 kt. As part of the debate on how well one would be able to estimate very low yields, there is a need for discussion of what restraints on allowed testing environment might be negotiated to give the monitoring side confidence that a new low-yield threshold was not being exceeded.

BASIC TERMINOLOGY AND DETECTION CAPABILITY

Three different types of seismic waves are described in this review—namely, body waves (specifically the *P*-wave), surface waves (specifically

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

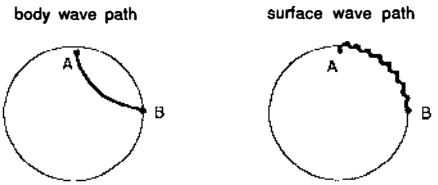

Figure 2 (*Left*) The path of a *P*-wave through the Earth, from a source at point A to a receiver at point B. This is a type of body wave, meaning that it travels through the Earth's deep interior. It is also a type of sound wave, traveling as a sequence of alternating compressions and rarefactions. An explosion at A may be recorded on a seismometer at B. (*Right*) This is a schematic for surface waves. A Rayleigh-wave path is shown, going from a source at point A to a receiver at point B. This wave travels around the Earth's surface and is conceptually similar to the set of ripples spreading away from a disturbance over the surface of a pond.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

the Rayleigh wave), and regional waves (specifically the *Lg*-wave). Each of these may be made the basis of a seismic method for yield estimation. *P*-waves and *Lg*-waves are important for assessing detection capability, since they are observable even for relatively small explosions.

A body wave is a seismic wave that propagates through the body of the spherical Earth [see Figure 2 (*left*)]. A surface wave travels a completely different path [Figure 2 (*right*)]. *P*-waves and Rayleigh waves (discovered by Lord Rayleigh in 1887) have long been used to estimate the size of seismic sources. Body waves and surface waves are said to be "teleseismic," meaning that they propagate to distances beyond 2000 km. In contrast, seismic waves such as *Lg* (sce Figure 3), are called "regional" not only because they do not propagate all over the globe, but also because they are different from region to region. The *Lg*-wave can be the most important of the regional waves for purposes of explosion monitoring because it is typically the largest wave observed in a seismogram at a regional distance. The *Lg*-wave has most of its energy trapped in the Earth's crust by a series of reflections at the crust-mantle interface (the "Moho"), very similar to the reflections of light occurring within an optical waveguide.

The basic data for seismic monitoring is a seismogram, a record of

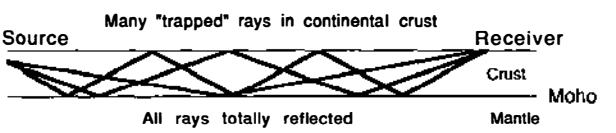

Figure 3 Schematic illustration of multiple total reflections, constituting the *Lg*-wave propagating in the low-velocity crustal waveguide.

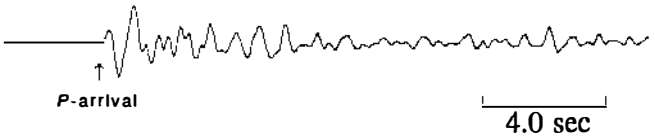

Figure 4 A teleseismic *P*-wave.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

ground motion at a particular location. In Figure 4 is a typical seismogram showing a *P*-wave (which is always the first wave to arrive, traveling faster than, for example, surface waves). This seismogram records the vertical motion of the ground, measured in Scotland (July 4, 1976), due to a Shagan River explosion. Note the time scale, indicating that the period of the *P*-wave is roughly 1 s. A seismologist would call this a short-period wave.

In order to obtain the seismic magnitude, one measures both the amplitude A of the maximum ground motion in the first few seconds and the period T of this motion. This leads into the concept of body-wave magnitude, symbolized as m_b, which is assigned by evaluating the formula

$$m_b = \log (A/T) + \text{distance correction},$$

where A is the maximum amplitude of the body wave, read off a seismogram at a particular distance and converted to ground motion by allowing for the instrumental gain; and T is the period (often around 1 s) of the wave at maximum trace amplitude.

The logarithm of ground motion is employed because of the wide range of sizes of observed seismic signals. The distance correction makes allowance for the distance between the receiver and the source of the seismic waves. In the seismogram above, for example, the correction is that appropriate for the distance between Scotland and Shagan River. The magnitude m_b has been used in the study of earthquakes since 1935. The scale is directly related to signal levels, and thus is a phenomenological quantity.

The teleseismic *P*-wave signal from a 1-kt underground nuclear explosion corresponds roughly to $m_b = 4$. Since signal strength is approximately proportional to yield and the magnitude scale is logarithmic, it follows that a 100-kt explosion has a signal strength of about $m_b = 6$, and that factors of 2 in yield (up or down) correspond to about 0.3 magnitude units (plus or minus).

In practice the relationship between magnitude and yield is complicated by naturally occurring variability in both the efficiency with which nuclear yield couples into a seismic signal at the shot point and the efficiency with which the seismic signal propagates to teleseismic distances. The degree of coupling depends on rock type at the shot point, on the amount of water saturation there, and on whether the explosive device is "tamped" (i.e.

well packed in the emplacement hole) or placed within some type of cavity. The efficiency of propagation to teleseismic distances is thought to depend basically on temperatures of the solid material in the upper layers of the Earth within which the seismic wave propagates for part of its path. The resulting effects on wave attenuation may be estimated for different propagation paths, using a wide variety of seismic data that may be quite separate from the data on the explosion under study.

For surface waves, the surface-wave magnitude M_s is assigned by

$$M_s = \log (A/T) + \text{distance correction}$$

where A is the amplitude of surface-wave ground motion, and T is the period at which this motion takes place. The main differences between this scale and the m_b scale stem from the two ways in which surface waves (see Figure 5) appear very different from body waves in a seismogram.

First, surface waves are dispersed, meaning that different frequency components travel at different speeds. The effect is to make a surface wave spread out in time over several minutes. The second way in which surface waves differ from body waves is that they are a long-period signal. The period T is typically in the wave range 18–20 s for the part of the well-dispersed wave train where amplitudes are at their maximum (where A is measured). Care must be taken to avoid measuring A at the so-called Airy phase, a poorly dispersed signal composed of a range of frequencies all arriving at the same time, creating a more impulsive motion.

It should be noted that in monitoring for small explosions under a new test ban regime, reliance would be placed not so much on teleseismic signals, propagating as illustrated in Figures 2, 4, and 5, but on regional waves as recorded at distances less than about 1000 km between source and receiver. There are several such seismic waves guided by layering in the crust and upper mantle—we have described *Lg* in Figure 3. Although for a given yield they are typically stronger than teleseismic waves, they are significantly harder to interpret because crustal and upper mantle

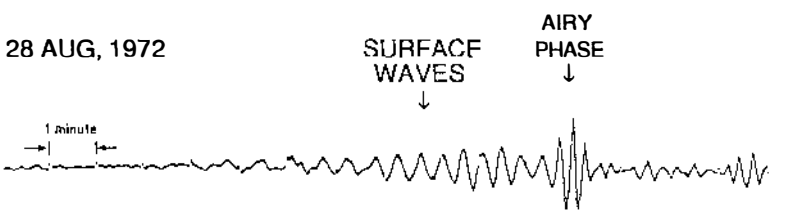

Figure 5 Example of a surface wave, teleseismically recorded at Eilat, Israel, from a large underground nuclear explosion ($m_b = 6.3$) at Novaya Zemlya.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

properties vary from region to region. Measures of regional-wave amplitude often show much more scatter than is the case for teleseismic observations. However, such scatter may be greatly reduced in specialized studies of a single region. This is important both for assessing detection and identification capabilities using regional waves and for purposes of yield estimation.

Some Examples of Detection Capability

The Earth's surface continuously undergoes small motions, stimulated by ocean tides, winds, surf, train and truck traffic, etc. Seismic signals from earthquakes and explosions can be detected to the extent that they are stronger than this background noise, which varies greatly at different sites and in different frequency bands. Deployment of seismometers is more effective if sites can be found that are very quiet.

The signals of interest in seismic monitoring have a relatively impulsive nature. It follows that explosion signals can be detected by comparing signal levels averaged over a short time (about 2 s or less) with levels averaged over a longer period (about 30 s) and continually updating the comparison. Such analysis has traditionally been done by eye, but it is now routinely automated.

With many different detectors, the next technical problem is recognizing an association between signals collected at different geographic locations. That is, the signals must be related to some presumed common source, such as an earthquake or an explosion (whose signals have been picked up by a network deployed over some region of the Earth or perhaps over the whole globe); and that source must be located. The basic method of location is the same as that by which we use our ears to tell from which direction a sound is coming. One analyzes the slight difference in times of arrival of signal at different places. For a good enough network and a strong enough source, the method has considerable redundancy. *P*-wave signals detected at four or more stations are usually sufficient to give a useful source location. A large underground explosion (150 kt) would be picked up at hundreds of seismometers around the world. A small explosion (1 kt) would be picked up on relatively few.

With many years of experience it has proven possible to assess the capabilities of a hypothetical network of detectors, given the location and sensitivity of each seismometer as well as a characterization of noise levels at each site in the network. A useful way to describe these capabilities, both actual and predicted, is by contouring the globe in magnitude units that give the threshold value of source strength above which a seismic source will be detected with high probability and at enough stations to be located by the network. Figure 6 shows the actual and the predicted value

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

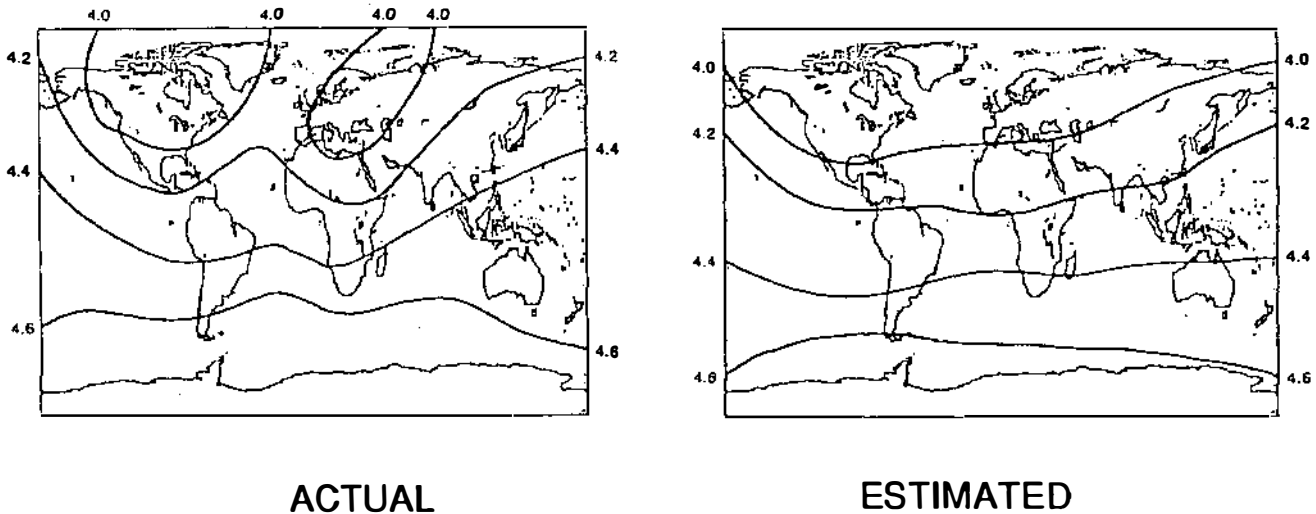

Figure 6 Contours showing detection capability of a particular seismic network of existing stations. For planning purposes, it is important that the estimated capability is in good agreement with the actual capability. Here, the differences are only about 0.1-m_b unit (Ringdal 1985).

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

of these contours, for a particular network of 115 globally distributed stations, requiring the quite stringent standard of 90% probability of detection (for events near the threshold level) at a minimum of four stations, at least one of which must be at a teleseismic distance.

Though Figure 6 shows a "globally distributed" network, it suffers from relatively poor coverage in the Southern Hemisphere. Thus, for this network, although the detection threshold has m_b in the range 3.9–4.3 in the Northern Hemisphere, the range is 4.2–4.8 in the south. To improve detection capability for sources in some particular part of the world, one would need to add stations located at distances favorable for picking up signals from that region.

Another practical element is introduced when one considers the capabilities of a group of seismometers in the same general region, all operated together as an "array," with central processing facilities that allow the multichannel signals to be filtered, delayed, and summed. Such an array is the seismologist's version of a phased array radar, permitting significant improvement in seismic signal-to-noise ratio as compared with this ratio for a single sensor. By adding the *P*-waves recorded at individual sensors (after an appropriate time shift to give all these first arrivals a common start time), the signal is enhanced if the *P*-wave first motion is coherent across the array. The relatively incoherent noise (which is apparent prior to the *P*-waves on each sensor) is suppressed. The outcome of such processing is often referred to as the "beam" formed by the array, which may be "steered" to "look" at different source regions by using the set of delays at each sensor appropriate for each source region, prior to signal summation. An array operated since about 1972 in southern Norway, known as NORSAR, with several tens of seismometers in a region about 200 km across, shows remarkable teleseismic detection capability for seismic sources in the USSR. The detection threshold for Semipalatinsk explosions, about 4000 km away, is about $m_b = 2.5$–3.0, and it is about 2.0–2.5 for the Caspian Sea area (Ringdal 1984). In June 1985, another seismic array became operational in southern Norway. Known as NORESS, it consists of 25 seismometer sites over an area about 3 km across, linked by optical fiber to a central processing facility. NORESS was designed for signals from sources at "regional" distances, and the detection threshold for this array is down to about $m_b = 2.0$–2.5 at 1500-km epicentral distance. At regional distances, the seismic waves *Pg*, *Pn*, *Sn*, and *Lg* are commonly seen, with *Lg* commonly being the strongest signal. [*Pg* propagates wholly within the crust; *Pn* and *Sn* travel in the top of the mantle, just below the Moho; and *Lg* is guided largely by the crustal layer (see Figure 3).] NORESS *P*-wave spectra extend up to about 18 Hz for quarry blasts about 1000 km away, and it has also been shown for a

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

single sensor that NORESS signals have important information in even higher frequency bands. Remarkably, for a magnitude 3.0 event 500 km away, the signal is about 20 dB above noise, all the way out to 50 Hz (Ringdal et al 1986).

From the above examples of detection capability, it is apparent that the research frontier for seismic monitoring is represented by events roughly in the range $2 \leq m_b \leq 4$, corresponding to fractions of a kiloton for typical explosions. We turn next to a characterization of different types of seismic sources.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

SEISMIC SOURCES

Most seismic signals from located events are caused by earthquakes, although chemical explosions are also recorded in large numbers. Nuclear explosions are relatively rare, and significant rockbursts even rarer. A monitoring network, however, will receive seismic signals created by all of these sources, and methods must be demonstrated not only to detect but also to identify with high confidence a clandestine nuclear test against the background of other seismic sources. In this section we review the basic properties of different types of seismic sources, contrasting them (where appropriate) with properties of nuclear explosions, in order to provide a physical basis for source identification.

Nuclear Explosions

For reference, we first comment on the production of seismic signals from the energy in a nuclear explosion—energy that is released in less than a microsecond. This energy produces a bubble of gas in which pressure and temperature rise steeply, reaching several million atmospheres and about 10^6 K within a few microseconds. A spherical shock wave expands outward from the bubble into the surrounding rock, crushing the rock as it travels and gradually weakening as it travels farther outward and away from the shot point (the point of emplacement of the nuclear device). As the hot gases expand and rock is vaporized near the shot point, a cavity is created. The size of this cavity is somewhat dependent on rock properties. While the cavity is forming, the shock wave continues to propagate away into the surrounding medium, eventually weakening to the point where rock no longer stays crushed but instead is merely compressed and returns (after a few cycles) essentially to its original state, thus becoming the propagating *P*-wave observed seismically. Because the first motion to occur in an explosion-generated *P*-wave is compressive, the first motion of the ground at the instant of the *P*-wave arrival is away from the source. It follows

that the first motion of the ground is always upward for a well-recorded explosion.

Earthquake Sources

Earthquake activity is a global phenomenon, though most of the larger earthquakes are concentrated in active tectonic regions along the edges of the Earth's continental and oceanic plates. In the USSR, most earthquakes are located in a few active regions along the border of the country. For example, in the Kurile Islands region, shallow earthquakes generally occur on the ocean side of the islands, whereas the deeper earthquakes occur beneath the islands and toward the USSR landmass. Large areas of the USSR have no earthquake activity and hence are referred to as *aseismic*. However, some activity may occur in these regions at magnitudes below $m_b = 3.0$.

Worldwide, about 7000 to 7500 earthquakes occur annually with $m_b \geq 4.0$, and about 7% of these occur in the USSR. About two thirds of Soviet earthquakes, however, occur under the ocean around the coast. The smaller the magnitude, the more earthquakes there are. Table 1 provides a summary of roughly how many earthquakes can be expected each year above a given magnitude.

Earthquakes and their associated seismograms have been studied on a quantitative basis for about 100 years. Since 1959, seismologists have engaged in a substantial research effort to discriminate between earthquakes and explosions by analyzing seismic signals. They have relied on a growing body of seismic observations to develop empirical source identification criteria, and on theoretical models to promote a better understanding of the underlying physical phenomena.

Table 1 Two estimates of the numbers of earthquakes occurring per year at or above various magnitude levels (m_b) of relevance for CTBT and low-yield TTBT monitoring

Seismic magnitude	Numbers of earthquakes, worldwide, estimated by: Ringdal (1985)	Lilwall & Douglas (1985)	Earthquakes in USSR land areas[a]
4.5	2600	2300	67
4.0	7400	7100	183
3.5	21,000	19,000	500
3.0	59,000	68,000	1400
2.5	166,000	209,000	4000

[a] About 2–3% of these events occur on Soviet continental territory (Bache 1985). This column (from US Congress 1988) gives numbers of earthquakes per year estimated for Soviet land areas.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

From such studies, three fundamental differences between nuclear explosions and earthquakes have been recognized:

1. differences in characteristic source location,
2. differences in source geometry,
3. differences in efficiency of seismic-wave generation at different wavelengths.

With regard to the first difference, many earthquakes occur deeper than 10 km, whereas the deepest underground nuclear explosion to date appears to have been around 2.5 km, and weapons tests under the 150-kt threshold all appear to be conducted at depths less than 1 km. Testing at depths much greater than current practice adds significantly to both expense and difficulty. Almost no holes, for any purpose, have been drilled to more than 10-km depth. The exceptions are few, well known, and are too substantial to hide. Because nuclear explosions are restricted to shallow depths, discrimination between many earthquakes and explosions on the basis of depth is possible in principle, although in practice the uncertainty in depth determination makes the division less clear. To compensate for the possibility of very deep emplacement of an explosion, the depth must be determined to be greater than 15 km with high confidence before the event can be identified unequivocally as an earthquake.

With regard to source geometry, the fundamental differences are due to how the seismic signals are generated. An underground nuclear explosion is a highly concentrated source of compressional seismic waves (*P*-waves), sent out with approximately the same strength in all directions. This type of signal occurs because the explosion products apply a relatively uniform pressure to the walls of the cavity created by the explosion. In constrast, an earthquake is generated as a result of massive rock failure that typically accomplishes a net shearing motion across a fault between two blocks. Because of this shearing motion, an earthquake can be thought of as radiating predominantly transverse motions, called *S*-waves, from all parts of the fault that rupture. Although *P*-waves from an earthquake are generated at a substantial fraction of the *S*-wave level, they are associated with a four-lobed radiation pattern of alternating compressions and rarefactions in the radiated first motions, rather than the relatively uniform pattern of *P*-wave compressions radiated in all directions from an explosion source. The idealized radiation patterns for *P*-waves are shown schematically in Figure 7.

With regard to the efficiency of seismic-wave generation, we note that earthquakes tend to have larger source dimensions than explosions because they involve a larger volume of rock. The bigger the source volume, the longer the wavelength of seismic waves set up by the source. As a result,

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

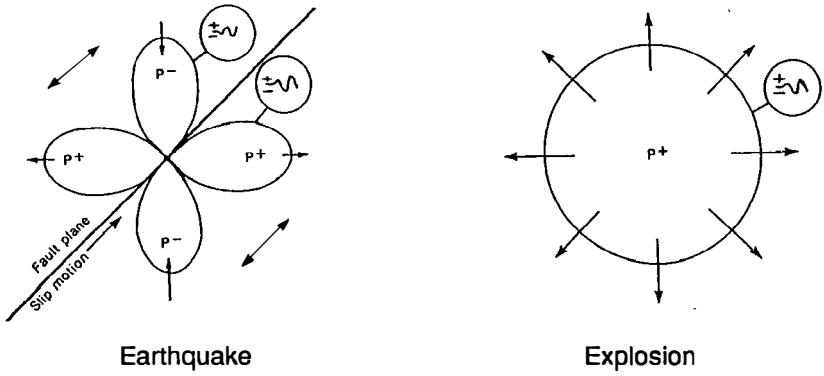

Figure 7 Radiation patterns of *P*-waves.

in comparing seismic signals of about the same strength from an earthquake and from an explosion, the earthquake signal is stronger at longer wavelengths and the explosion signal is stronger at short wavelengths. Instead of stating the outcome in terms of wavelength, an equivalent description can be given in terms of seismic-wave frequency. Thus, explosions, which entail a smaller but more intense source region, typically send out stronger signals at high frequency (see Figure 8). However, exceptions have been observed, and the distinction may diminish or even reverse for small events. The difference in frequency content of the respective signals, particularly at high frequencies, is a topic of active research, and as new types of seismic data become available (for example, data from within the USSR) it may be expected that methods of identification will be found that work at smaller magnitude levels. In fact, the success of identification methods for distinguishing between nuclear explosions and earthquakes is such that the remaining fundamental difficulties (which arise only at the lower yields) are in several respects less than the difficulties of distinguishing seismically between nuclear explosions and chemical explosions.

Chemical Explosions

Chemical explosions are used routinely in the mining and construction industries and also occur in military programs and on nuclear test sites. In general, from the seismic monitoring perspective, a chemical explosion can be considered to be a small spherical source of energy very similar to a nuclear explosion. The magnitudes represented by chemical explosions are below $m_b = 4.0$, with few exceptions (which are well known). The fact that large numbers of chemical explosions in the yield range 0.001–0.01 kt

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

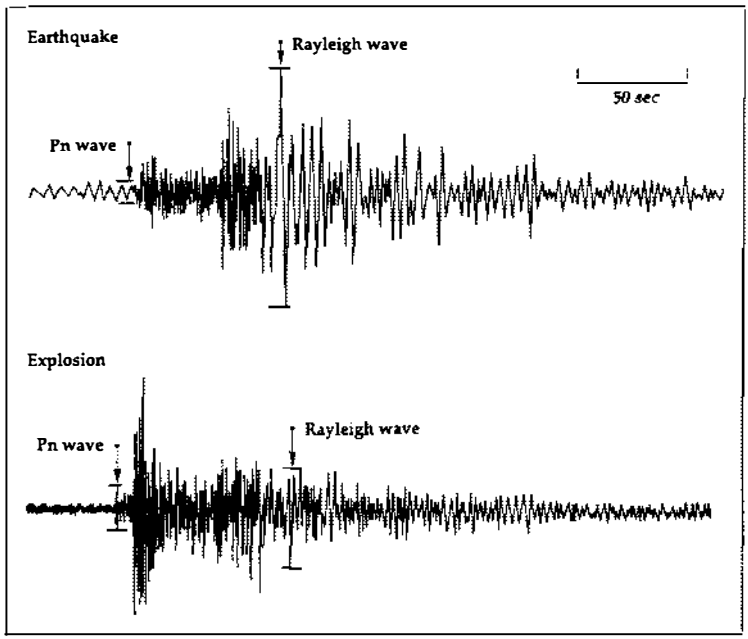

Figure 8 Seismograms for an earthquake and an explosion located near each other at the Nevada Test Site. Note the larger size of the (high-frequency) *Pn*-wave for the explosion, and the larger size of the (low-frequency) Rayleigh wave for the earthquake (figure from Jarpe 1989).

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

are detected and located seismically is a testament to the capability of seismic networks to work with signals below $m_b = 3.0$.

There is little summary information available on the number and location of chemical explosions in the USSR, although it appears that useful summaries could be prepared from currently available seismic data. In the US, chemical explosions of yield around 0.2 kt are common at about 20 mines. At each of these special locations, tens of such explosions may occur each year. Presuming that similar operations are mounted in the USSR, this activity is clearly a challenge when monitoring for nuclear explosions with yields below around 1 kt. It is also a challenge when considering the possibility that a nuclear-testing nation might try to muffle a larger nuclear explosion (say, around 5 kt), so that its seismic signals resembled those of a much smaller (say, around 0.1 kt) chemical explosion.

Progress in identifying chemical explosions can be based on three approaches:

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

1. Almost all chemical explosions for industrial purposes listed with yields at 0.1 kt or greater are in fact a series of more than 100 small explosions spaced a few meters apart and fired with small time delays between individual detonations. An effect of such "ripple firing" is to generate seismic signals that are like those of a small earthquake, in that both these types of sources (ripple-fired chemical explosion, and small earthquake) are spatially extended and thus lose some of the characteristics of a highly concentrated source, such as a nuclear explosion. Recent research has shown the importance of acquiring seismic data at high frequency (30–50 Hz); indeed, such data suggest the existence of a distinctive signature for ripple-fired explosions (Baumgardt & Ziegler 1988, Hedlin et al 1989).
2. The rare chemical explosions above about 0.5 kt that are not ripple fired can be expected to result in substantial ground deformation (e.g. cratering). Such surface effects, together with the absence of a radiochemical signal, would indicate a chemical rather than a nuclear explosion. The basis for this discriminant is that chemical explosions are typically very shallow.
3. To the extent that remote methods of monitoring chemical explosions are deemed inadequate (for example, in a low-yield threshold or comprehensive test ban regime), solutions could be sought in requiring prior announcement of certain types of chemical explosion. Note that chemical explosions above 0.1 kt occur routinely only at a limited number of sites. Some type of minimal on-site inspection and/or a program of in-country radiochemical monitoring could also make a contribution.

For an cxplosion that is well coupled into the ground (a "tamped" explosion), a magnitude of $m_b = 2.0$ can be achieved with a yield of about 0.01 kt. At this level, a monitoring program would confront perhaps 1000 to 10,000 chemical explosions per year. Given the difficulty of discriminating between chemical and nuclear explosions, it is clear that seismic monitoring fails at some levels of low but still detectable yield. Thus, a truly comprehensive ban on all nuclear testing could not be strictly verified seismically. However, recognizing that relatively few chemical explosions occur at the upper end of this m_b range (2.0–4.0), where discriminants between chemical and nuclear explosions do exist, data recorded within the USSR can clearly contribute substantially to an improved monitoring capability.

Rockbursts

In underground mining involving tunneling activities, the rock face in the deeper tunnels may occasionally burst suddenly into the tunnel. This is

referred to as a rockburst and results from the difference between the low pressure existing within the tunnel and the much greater pressure within the surrounding rock.

In terms of magnitude, rockbursts are all small. They occur over very restricted regions in the Earth and generally have a seismic magnitude of less than $m_b = 4.0$.

The source mechanisms of rockbursts are very similar to those of small earthquakes. In particular, the direction of the first seismic motion from a rockburst will have a pattern similar to that observed for earthquakes. Therefore, for the seismic identification problem, rockbursts can be considered small earthquakes that occur at very shallow depths.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

IDENTIFICATION METHODS

Over the years, a number of identification methods have been shown to be fairly robust. Some of these methods perform the identification process by identifying certain earthquakes as being earthquakes (but not identifying explosions as being explosions). Other methods identify certain earthquakes as being earthquakes and certain explosions as being explosions. The identification process is therefore a type of winnowing operation.

Location

The principal identification method is based on the location of a detected seismic source.

If the epicenter is determined to be in an oceanic area but no explosion hydroacoustic signals are recorded, then the event is identified as an earthquake. Large numbers of seismic events can routinely be identified in this way, since so much of the Earth's seismic activity occurs beneath or adjacent to ocean basins. If the location is determined to be in a land region, then in certain cases the event can still be identified as an earthquake on the basis of location alone [for example, if the site is clearly not suitable for nuclear explosions (such as near population centers) or if there is no evidence of human activity in the area].

Depth

With the exception of epicenter interpretation, interpretation of the seismic source depth is probably the most useful identification technique. A seismic event can be identified as an earthquake if its depth is determined with high confidence to be below 15 km.

The procedure for determining source depth is a part of the process of finding the event location using the arrival time of four or more *P*-wave signals. A depth estimate can be obtained by measuring the time difference

between the first arriving *P*-wave energy and the arrival time of signals reflected from the Earth's surface above the source.

An advantage of depth as an identification method is that it is not dependent on magnitude—it will work for small events as well as large ones, provided that the basic data are of adequate signal quality. It will not alone, however, distinguish between chemical and nuclear explosions unless the nuclear explosion is relatively large and deep; nor will it distinguish between underground nuclear explosions and earthquakes unless the earthquakes are sufficiently deep.

$M_s : m_b$

Underground nuclear explosions generate signals that tend to have surface-wave magnitudes M_s and body-wave magnitudes m_b that differ systematically from earthquake signals. This is basically an expression of the efficiency of explosions in exciting body waves (short period) and of earthquakes in exciting surface waves (long period). The phenomenon is often apparent in the original seismograms (see Figure 8), and an example is shown in terms of an $M_s : m_b$ diagram in Figure 9. These diagrams can be thought of as separating the population of explosions from that of earthquakes. For any event that is clearly in one population or another, the event is identified. For any event that lies between the two populations (as occasionally happens, particularly with lower quality seismic data for explosions at the Nevada Test Site), this method does not provide reliable identification.

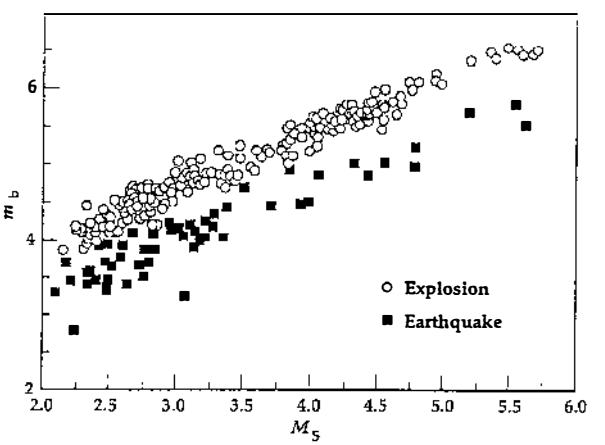

Figure 9 $M_s : m_b$ diagram for seismic sources at or near the Nevada Test Site. Note the separation into two populations (figure from Jarpe 1989).

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

To use this identification method, both m_b and M_s values are required for the event. This is no problem for the larger events, but for smaller events (below $m_b = 4.5$) it can be very difficult to detect long-period surface waves using external stations alone. This difficulty arises particularly for explosions, with the result that the $M_s : m_b$ method intrinsically works better for identifying small earthquakes than it does for identifying small explosions. Internal stations would provide important additional capability for obtaining M_s values of events down to $m_b = 3.0$ and perhaps below.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

Spectral Methods

The basis of the success of $M_s : m_b$ diagrams is that the ratio of low-frequency waves to high-frequency waves is typically different for earthquakes and explosions. Thus, M_s is a measure of signal strength at around 20 s (frequency 0.05 Hz), and m_b is a measure of signal strength at around 1 s (frequency 1 Hz). As a way of exploiting such differences, $M_s : m_b$ diagrams are quite crude compared with methods that use a more complete characterization of the frequency content of seismic signals.

What then can be learned from analysis of earthquake and explosion signals across the whole spectrum of frequencies that they contain? This question drives much of the current research and development efforts in seismic monitoring.

Because $M_s : m_b$ diagrams work so well when surface waves are large enough to be measured, the main contribution required of more sophisticated analysis is that it make better use of the information contained in short-period *P*-wave signals. This offers the prospect of improved identification capabilities just where they are most needed, namely for smaller events. Thus, instead of boiling down the *P*-wave signals at all stations simply to a network m_b value, one can measure the amplitude at a variety of frequencies and seek a discriminant that works on systematic differences in the way that earthquake and explosion signals vary with frequency. One such procedure is the variable frequency magnitude (VFM) method, which measures the short-period body-wave signal strength from seismograms filtered to pass energy at two different frequencies, f_1 and f_2. Discrimination is based on a comparison of $m_b(f_1)$ and $m_b(f_2)$, which in many cases shows a clear separation of earthquake and explosion populations (Stevens & Day 1985).

High-Frequency Signals

The use of high-frequency signals in seismic monitoring is strongly linked to the question of what can be learned from seismic stations in the USSR whose data are made available to the US. Such in-country or internal

stations were considered in CTBT negotiations in the late 1950s and early 1960s, and the technical community concerned with monitoring in those years was well aware that seismic signals up to several tens of hertz could be observed to propagate from explosions out to distances of several hundred kilometers—what we have referred to previously as "regional distances."

However, this early interest in regional waves lessened once the Limited Test Ban Treaty of 1963 was signed, because it was then recognized that subsequent programs of underground nuclear explosions conducted by different countries could be monitored teleseismically. At teleseismic distances, seismic body-wave signals are usually simpler, are usually not seen with frequency content above 10 Hz, and have indeed proven adequate for most purposes of current seismic monitoring, even under the treaty restrictions on underground nuclear testing that have developed since 1963. It is in the context of considering further restraints on testing, such as a CTBT or LYTTBT, that the seismic monitoring community is required to evaluate high-frequency/regional seismic signals. In such a hypothetical testing regime, the greatest challenge to monitoring will come from very small nuclear explosions, and for them the most favorable signal-to-noise ratios will often be at high frequency (Evernden et al 1986). Internal stations, producing data of high quality and at quiet sites, will be essential in addressing the monitoring issues associated with such events.

The claim that high-frequency seismic data from internal stations in the USSR can lead to great improvements in detection capability has been supported by the data recorded within that country since 1987. Nongovernmental agreements have allowed US seismologists to deploy sophisticated monitoring equipment at sites within Soviet Central Asia (Given et al 1990, Hansen et al 1989).

The daily recording of very small seismic events, made possible by improvements in detection capability, focuses attention on the discrimination problem for events in the magnitude range $2.0 \leq m_b \leq 4.0$. In practice, the main use of in-country high-frequency signals (i.e. above 10 Hz) in improving identification capability may lie in interpretation of the improved locations, including depth estimates (Ruud et al 1988, Thurber et al 1989).

TEST BAN EVASION

Several methods havc been suggested by which a low-threshold test ban might be evaded. These include testing behind the Sun or in deep space; detonating a series of explosions to simulate an earthquake; testing during or soon after an earthquake; testing in large underground cavities, in

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

nonspherical cavities, or in low-coupling material such as deposits of dry alluvium; and masking a test with a large, legitimate industrial explosion.

Although the majority of proposed evasion scenarios have been shown to be readily defeated, a few concepts have evoked serious concern and analysis. We describe them below, in what we believe roughly to be the order of increasing effectiveness.

Testing Behind the Sun or in Deep Space

The concern is that one or more vehicles sent into space could detonate a nuclear device and record the testing information to relay back to Earth. Some feel that such a testing scenario is both technically and economically feasible. Others believe that the technical and economic resources required, when viewed in terms of the uncertainty and risk of discovery, would greatly exceed the utility of any information that could be obtained. [Note the logistical effort implied by the fact that testing operations currently engage 8000 employees at the Nevada Test Site, plus many more of the 10,000 employees at each of the three US national weapons laboratories (Los Alamos, Livermore, and Sandia); and that many thousands of people were needed to execute the US atmospheric test program in the Pacific up to 1963.] A nuclear test explosion in space would be an unambiguous violation of several treaties, making discovery very costly to the cheater. If testing can be demonstrated on technical grounds to be a concern, the risk could be addressed, although at high cost, by deploying satellites to orbit the Sun. Alternatively, the risk could be addressed politically by negotiating an agreement to conduct simple inspections of the rare vehicles that go into deep space.

Simulating an Earthquake

It has been suggested that a series of nuclear explosions could be sequentially detonated over a period of a few seconds to mimic the seismic signal created by a naturally occurring earthquake. However, it would be a poor mimic, working only if the *P*-wave amplitude, over just one cycle of motion, were measured at stations within a certain distance range and if reliance were placed mainly in the $M_s:m_b$ discriminant. Consequently, a network that records over a variety of distances and/or records *P*-wave amplitudes measured over several cycles could be used to monitor against this scenario.

Testing During an Earthquake

If a small nuclear weapons test were to be detonated shortly after a nearby naturally occurring earthquake, the signals of the explosion would to some extent be obscured by earthquake signals. The logistical difficulties of

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

holding a test in readiness for months or years would be substantial, and strong nerves and prompt good judgment would be required to identify the suitability of a candidate earthquake within less than a minute of its occurrence. (Note that seismologists currently have no reliable techniques for the short-term prediction of the time, location, and size of earthquakes, and this limitation is unlikely to be overcome in the near future.)

The feasibility of this scenario is further reduced by using networks that record high-frequency waves. The obscuring effects of an earthquake are much reduced in a high-frequency record. For this reason, and for the logistical difficulties, the hide-in-earthquake scenario is no longer considered a credible evasion threat.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

Testing in Low-Coupling Materials

The fraction of energy from a nuclear explosion that is converted into seismic energy (and thus into possibly detectable signals) depends on the type of rock in which the explosion occurs. Low-coupling materials such as dry porous alluvium have air-filled pore spaces that absorb much of the explosive energy. This has led to concern that a monitoring network could be evaded by detonating an explosion in such material. However, the opportunities for such evasion are thought to be limited in the Soviet Union, where no great thicknesses of dry alluvium are known to exist. This contrasts with the geology of the Nevada Test Site, which, with its unconsolidated sediments and low water table, has some of the most unusual geology in the world. Estimates of the maximum thickness of alluvium in the USSR indicate that it would be sufficient to muffle explosions only up to 1 or 2 kt. In addition, because alluvium is easily disturbed, an explosion could create a subsidence crater or other telltale evidence.

Masking a Test With a Chemical Explosion

Unlike an earthquake, large chemical explosions, such as those used routinely in mining and construction, could be timed to coincide with a clandestine nuclear test. If done in combination with cavity decoupling, to hide a small nuclear test as discussed below, this scenario presents perhaps the most serious challenge to a monitoring network.

Cavity Decoupling

If a nuclear explosion is set off in a sufficiently large underground cavity, it will emit seismic waves that are much smaller than those from the same-size explosion detonated in a tamped underground test. The physical principles underlying this scheme, called cavity decoupling, have been verified at small yields. It appears that opportunities exist within the USSR

and other countries to construct underground cavities suitable for so muffling low-yield explosions. The easiest method would probably be via solution mining in a salt dome. Consequently the capability to conduct clandestine decoupled nuclear tests determines the yield threshold below which treaty verification, by seismic means alone, is no longer possible with high confidence.

An underground nuclear explosion creates seismic waves with a broad range of frequencies. For purposes of seismic detection and discrimination of small explosions, frequencies from about 1 Hz to as high as 30 or 50 Hz may be important.

For the lower end of this frequency range, the amplitude of the waves created by the explosion is approximately proportional to the total amount of new cavity volume created by the explosion. A standard test is conducted in a hole whose initial volume is negligible compared with its posttest volume. Because the initial hole is small, the rock surrounding the explosion is strained at high levels and flows plastically. This flow results in large displacements just outside the cavity, leading to a large net increase in the volume of rock in the vicinity of the shot point, and hence to efficient generation of seismic waves.

If, on the other hand, the explosion occurs in a hole of much greater initial volume, the explosive stresses at the cavity wall will be much smaller, resulting in less flow of rock and hence less volumetric expansion and reduced excitation of seismic waves. If the initial hole is so large that the stresses in the surrounding rock never exceed the limit at which rock flow begins to occur, the seismic coupling is minimized. Further increase of the emplacement hole size will not further reduce coupling at low frequencies, and the explosion is said to be "fully decoupled." For strong salt or granite, the cavity radius required for full decoupling has been estimated as somewhat greater than 20 m times the cube root of the yield (in kilotons). For weak salt it is about 50% greater. However, the signal strength is not nearly so strongly reduced by decoupling at frequencies above that associated with resonances of the internal surface surrounding the shot point at which rock flow ceases. In practice, the frequency above which decoupling is likely to be substantially less effective is around 10 to 20 Hz, divided by the cube root of the yield.

From the above discussion, we can distinguish between the efficiency of full decoupling at low frequencies and the efficiency at high frequencies (Larson 1985). At low frequencies, the amount by which seismic-wave amplitudes are reduced by full decoupling is about a factor of 70, described equivalently as a reduction by 1.85 m_b units. At high frequencies, the reduction is about a factor of 7 (i.e. 0.85 m_b units).

Of course, a thorough discussion of decoupling as an evasion scenario

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

would have to include a number of nonseismological considerations. These include the military significance of being able to carry out nuclear tests up to various different yield levels (e.g. 0.1, 1, or 10 kt), and the political consequences if a clandestine test program were uncovered. Technical considerations include methods of (clandestine) cavity construction, and the capabilities of nonseismological surveillance techniques.

An impression of the scale of effort required to fully decouple a 5-kt underground nuclear explosion is conveyed in Figure 10. Such an explosion would still generate detectable seismic signals on a well-designed in-country network and would thus be likely to attract some attention.

SUMMARY OF IDENTIFICATION CAPABILITY

It is difficult to reach precise but general conclusions on what future low-yield levels could be monitored because much would depend on the degree

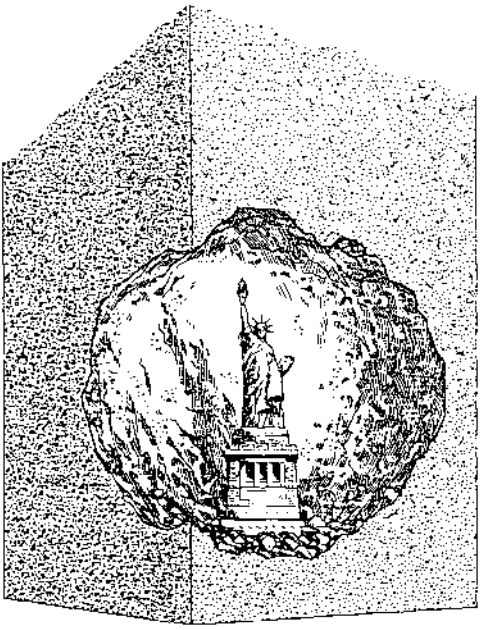

Figure 10 To fully decouple a 5-kt explosion in salt (the easiest medium in which to construct a large cavity), a spherical cavity with a diameter of at least 85 m would be required. The Statue of Liberty (height 74 m, including pedestal) is shown for scale (US Congress 1988).

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

of effort put into new seismic networks and new data analysis. It would depend, too, on judgments about the level of effort that might be put into clandestine nuclear testing.

The identification capability of any network will always be poorer than the detection capability for that network. Experience with teleseismic signals indicates that the difference amounts to about 0.5 m_b units—that is, a detection threshold of about 3.5 indicates an identification threshold of about 4.0 (Hannon 1985). However, for regional signals from events with magnitude less than $m_b = 3.5$, the difference may be larger than 0.5 m_b. A summary statement conveyed in a May 1988 Office of Technology Assessment Report (US Congress 1988) was as follows:

> . . . there appears to be agreement that, with internal stations that detect down to m_b 2.0–2.5, identification can be accomplished in the U.S.S.R. down to *at least* as low as m_b 3.5. This cautious identification threshold is currently set by the uncertainty associated with identifying routine chemical explosions that occur below this level. Many experts claim that this identification threshold is too cautious and that with an internal network, identification could be done with high confidence down to m_b 3.0.

This cautious identification threshold corresponds to a subkiloton yield for a tamped explosion. But if we are doubly cautious and go on to interpret it in terms of a substantially decoupled shot, the yield would be about 5 kt. By this argument, which can hardly be refuted on seismological grounds alone until one can build confidence in the routine achievability of an identification threshold lower than $m_b = 3.5$, the bottom line is a capability to monitor above about 5 kt.

FUTURE DIRECTIONS

Once the yield estimation issues related to the 150-kt Threshold Test Ban Treaty are politically resolved, it is required under the declared agenda of current US-USSR negotiations that attention turn to "further limitations on nuclear testing." Given the difficulty, if not the impossibility, of monitoring a truly comprehensive test ban, it appears from the perspective of verification that there is merit in reducing the allowed yield threshold in stages. Quoting again from the 1988 Office of Technology Assessment Report, which has guided much of the presentation of material in this review, such a staged approach "would begin with a limit that can be monitored with high confidence using current methods, but would establish the verification network for the desired lowest level. The threshold would then be lowered as information, experience, and confidence increase" (US Congress 1988).

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

ACKNOWLEDGMENTS

We thank Karen Fischer, Bill Haxby, and Won-Young Kim for reviews of the manuscript, as well as our many colleagues whose work we have drawn upon extensively, especially Steve Day, Frode Ringdal, Lynn Sykes, and Gregory van der Vink. We acknowledge a grant from the MacArthur Foundation to the School of International and Public Affairs at Columbia. This article is Lamont-Doherty Geological Observatory Contribution No. 4578.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

Literature Cited

Bache, T. C. 1985. Seismic identification of small events. See Larson 1985, pp. V-101–V-142

Baumgardt, D. R., Ziegler, K. A. 1988. Spectral evidence for source multiplicity in explosions: applications to regional discrimination of earthquakes and explosions. *Bull. Seismol. Soc. Am.* 78: 1773–95

Batzel, R. 1987. In *US Congress, Senate Foreign Relations Committee, Hearing 100–115*, pp. 220–21. Washington, DC: US Gov. Print. Off.

Evernden, J. F., Archambeau, C. B., Cranswick, E. 1986. An evaluation of seismic decoupling and underground nuclear test monitoring using high-frequency seismic data. *Rev. Geophys.* 24: 143–215

Given, H. K., Tarasov, N. T., Zhuravlev, V., Vernon, F. L., Berger, J., Nersesov, I. L. 1990. High-frequency seismic observations in eastern Kazakhstan, USSR, with emphasis on chemical explosion experiments. *J. Geophys. Res.* 95: 295–307

Hannon, W. J. 1985. Seismic verification of a comprehensive test ban. *Science* 227: 251–57

Hansen, R. A., Ringdal, F., Richards, P. G. 1989. Analysis of IRIS data for Soviet nuclear explosions. In *NORSAR Semiannual Technical Summary, 1 Oct 1988–31 Mar 1989*, pp. 121–40. Kjeller, Nor: NTNF/NORSAR

Heckrotte, W. 1988. On-site inspection to check compliance. In *Nuclear Weapons Tests: Prohibition or Limitation*, ed. J. Goldblat, D. Cox, pp. 248–49. Oxford: Oxford Univ. Press. 423 pp.

Hedlin, M., Minster, J. B., Orcutt, J. A. 1989. The time-frequency characteristics of quarry blasts and calibration explosions recorded in Kazakhstan U.S.S.R. *Geophys. J. Int.* 99: 109–21

Jarpe, S. P. 1989. Lawrence Livermore National Laboratory Nevada Test Site Regional Seismic Network. *Brochure LLL-TB-100*, Lawrence Livermore Natl. Lab., Livermore, Calif.

Larson, D. B., ed. 1985. *Proceedings of DOE-Sponsored Cavity Decoupling Workshop, Pajaro Dunes, Calif.* Washington, DC: US Dep. Energy

Lilwall, R. C., Douglas, A. 1985. Global seismicity in terms of short-period magnitude m_b based on individual station magnitude-frequency distributions. *Rep. O 23/84*, UK At. Weapons Res. Establ. London: Her Majesty's Stationery Off.

Richards, P. G. 1988. Seismic methods for verifying test ban treaties. In *Nuclear Arms Technologies in the 1990's. AIP Conf. Proc. No. 178*, ed. D. Schroeer, D. Hafemeister, pp. 54–108. New York: Am. Inst. Phys.

Ringdal, F. 1984. Teleseismic detection at high frequencies using NORSAR data. In *NORSAR Semiannual Technical Summary, 1 Apr–30 Sept 1984*, pp. 54–62. Kjeller, Nor: NTNF/NORSAR. 69 pp.

Ringdal, F. 1985. Study of magnitudes, seismicity, and earthquake detectability using a global network. In *The VELA Program: A Twenty-Five Year Review of Basic Research*, ed. A. U. Kerr, pp. 611–24. Washington, DC: Exec. Graph. Serv. 964 pp.

Ringdal, F., Hokland, B. Kr., Kværna, T. 1986. Initial results from the NORESS high frequency seismic element, (HFSE). In *NORSAR Semiannual Technical Summary, 1 Oct 1985–31 Mar 1986*, pp. 31–39. Kjeller, Nor: NTNF/NORSAR. 93 pp.

Ruud, B. O., Husebye, E. S., Ingate, S. F., Christoffersson, A. 1988. Event location at any distance using seismic data from a single, three-component station. *Bull. Seismol. Soc. Am.* 78: 308–25

Seaborg, G. T. 1981. *Kennedy, Khrushchev and the Test Ban.* Berkeley: Univ. Calif. Press. 320 pp.

Scheimer, J. F., Borg, I. Y. 1984. Deep seismic sounding with nuclear explosives in the Soviet Union. *Science* 226: 787–92

Stevens, J. L., Day, S. M. 1985. The physical basis of $m_b : M_s$ and variable frequency magnitude methods for earthquake/explosion discrimination. *J. Geophys. Res.* 90: 3009–20

Sykes, L. R., Ruggi, S. 1989. Soviet nuclear testing. In *Nuclear Weapons Databook*, Vol. 4, *Soviet Nuclear Weapons*, ed. T. B. Cochran, W. A. Arkin, R. S. Norris, J. I. Sands, pp. 332–82. Cambridge, Mass: Ballinger

Thorn, R. N., Westervelt, D. R. 1987. Hydronuclear experiments. *Rep. LA-10902-MS, UC-2*, Los Alamos Natl. Lab., Los Alamos, N. Mex.

Thurber, C., Given, H. K., Berger, J. 1989. Regional seismic event location with a sparse network: application to eastern Kazakhstan, USSR. *J. Geophys. Res.* 94: 17,767–80

US Congress 1988. *Seismic Verification of Nuclear Testing Treaties. OTA-ISC-361*, Off. Technol. Assess. Washington, DC: US Gov. Print. Off. 139 pp.

York, H. 1987. *Making Weapons, Talking Peace*. New York: Basic Books. 359 pp.

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.

Annual Review of Earth and Planetary Sciences
Volume 18, 1990

CONTENTS

WHAT IS A SPACE SCIENTIST? AN AUTOBIOGRAPHICAL EXAMPLE, *James A. Van Allen* 1

THE VALLES/TOLEDO CALDERA COMPLEX, JEMEZ VOLCANIC FIELD, NEW MEXICO, *Grant Heiken, Fraser Goff, Jamie N. Gardner, W. S. Baldridge, J. B. Hulen, Dennis L. Nielson, and David Vaniman* 27

CRITICAL TAPER MODEL OF FOLD-AND-THRUST BELTS AND ACCRETIONARY WEDGES, *F. A. Dahlen* 55

LATE PRECAMBRIAN AND CAMBRIAN SOFT-BODIED FAUNAS, *S. Conway Morris* 101

GEOLOGICAL AND BIOLOGICAL CONSEQUENCES OF GIANT IMPACTS, *Digby J. McLaren and Wayne D. Goodfellow* 123

SEAFLOOR HYDROTHERMAL ACTIVITY: BLACK SMOKER CHEMISTRY AND CHIMNEYS, *K. L. Von Damm* 173

FORMATION OF THE EARTH, *George W. Wetherill* 205

SEISMIC DISCRIMINATION OF NUCLEAR EXPLOSIONS, *Paul G. Richards and John Zavales* 257

BRINE MIGRATIONS ACROSS NORTH AMERICA—THE PLATE TECTONICS OF GROUNDWATER, *Craig M. Bethke and Stephen Marshak* 287

THE ORIGIN AND EARLY EVOLUTION OF LIFE ON EARTH, *J. Oró, Stanley L. Miller, and Antonio Lazcano* 317

THE NATURE OF THE EARTH'S CORE, *Raymond Jeanloz* 357

SYNCHROTRON RADIATION: APPLICATIONS IN THE EARTH SCIENCES, *W. A. Bassett and G. E. Brown, Jr.* 387

INDEXES
Subject Index 449
Cumulative Index of Contributing Authors, Volumes 1–18 461
Cumulative Index of Chapter Titles, Volumes 1–18 464

Annu. Rev. Earth Planet. Sci. 1990.18:257-286. Downloaded from www.annualreviews.org
Access provided by University of New Mexico (UNM) on 12/23/23. For personal use only.